

BGP, QoS & Capacity Planning — Technical Specification (Rev 4.1)

Core Network Engineering — Internal Reference

1. Peering Overview

This document defines eBGP peering, QoS marking, and capacity planning parameters for AS-64500 edge routers across three regional edge sites: EDGE-1, EDGE-2, and EDGE-3. All sessions use multihop TTL of 2 unless otherwise noted, and MD5 authentication is mandatory except for IX route-server peers.

2. Peer Table

Peer	Remote ASN	Site	Session Type	Keepalive/Hold (s)	Max Prefixes
UpstreamA	AS-7018	EDGE-1	eBGP Transit	10 / 30	900,000
UpstreamB	AS-3356	EDGE-1	eBGP Transit	10 / 30	950,000
UpstreamC	AS-1299	EDGE-2	eBGP Transit	10 / 30	920,000
IX-PeerX	AS-24482	EDGE-1	eBGP Peering (IX)	3 / 9	50,000
IX-PeerY	AS-4788	EDGE-2	eBGP Peering (IX)	3 / 9	45,000
IX-PeerZ	AS-9902	EDGE-3	eBGP Peering (IX)	3 / 9	38,000
CustomerZ	AS-65010	EDGE-1	eBGP Customer	5 / 15	1,000
CustomerW	AS-65022	EDGE-3	eBGP Customer	5 / 15	1,500

3. Sample Configuration — UpstreamA (EDGE-1)

```
router bgp 64500
  neighbor 198.51.100.1 remote-as 7018
  neighbor 198.51.100.1 timers 10 30
  neighbor 198.51.100.1 maximum-prefix 900000 90
  neighbor 198.51.100.1 route-map RM-UPA-IN in
  !
  route-map RM-UPA-IN permit 10
    set local-preference 150
    set community 64500:100 additive
```

4. Sample Configuration — UpstreamC (EDGE-2)

```
router bgp 64500
  neighbor 198.51.100.5 remote-as 1299
  neighbor 198.51.100.5 timers 10 30
  neighbor 198.51.100.5 maximum-prefix 920000 90
  neighbor 198.51.100.5 route-map RM-UPC-IN in
  !
  route-map RM-UPC-IN permit 10
    set local-preference 130
    set community 64500:100 additive
```

5. QoS Marking Policy

Traffic is classified into three QoS classes at ingress based on DSCP marking: Voice (EF, DSCP 46) receives strict-priority queuing up to 10% of interface bandwidth; Business-Critical (AF31, DSCP 26) receives guaranteed minimum bandwidth of 30% with the ability to burst higher when capacity allows; Best-Effort (DSCP 0) receives the remaining bandwidth with no guarantee and is the first class dropped under congestion. Any traffic arriving without a recognized DSCP marking is treated as Best-Effort by default, not as an error condition.

6. Community Tag Policy

Routes from customers are tagged 64500:200 and advertised to all upstreams and IX peers. Routes from IX peers are tagged 64500:300 and are advertised only to customers, never to upstreams or other IX peers. Routes from upstreams are tagged 64500:100 and are customer-facing only, never re-advertised to upstreams or IX peers. If a route matches more than one condition, the more restrictive tag takes precedence, where restrictiveness ranks IX (300) as most restrictive, followed by upstream (100), followed by customer (200) as least restrictive.

7. Capacity Planning Formulas

Required uplink capacity: **Required_Mbps = (Peak_Concurrent_Users × Avg_Session_Kbps) / 1000 × Overprovision_Factor**. Standard Overprovision_Factor is 1.4. Avg_Session_Kbps is currently 220 Kbps per user, measured quarterly.

Redundancy headroom: **Headroom_Percent = ((Current_Uplink_Mbps – Required_Mbps) / Current_Uplink_Mbps) × 100**. Sites with Headroom_Percent below 15% are flagged for capacity upgrade planning in the following quarter.

8. Site Capacity Table

Site	Peak Concurrent Users	Current Uplink (Gbps)
EDGE-1	26,000	10
EDGE-2	31,000	10
EDGE-3	14,500	5

9. Troubleshooting — Known Error Codes

Error	Session Type	Typical Cause	Action
Hold Timer Expired	IX Peer	IX route-server maintenance	Monitor; usually self-resolves
Hold Timer Expired	Transit	Path MTU or link issue	Check transport path before escalating
Max Prefix Limit Exceeded	Transit	Upstream route leak or growth	Manual re-enable after confirming count with upstream
Max Prefix Limit Exceeded	Customer	Customer-side misconfiguration	Contact customer NOC directly